

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	C	C	A	B	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,D	A,C	A,B,C,D	A,B,D	C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.50	4.00	1.72 to 1.73	2.00	0.00	1.50

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	C	A	C	C	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	B,C,D	B,D	A,B,D	A,C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	7.00	2.00	135.00	6.00	4.00	8.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	A	A	A	C	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	B,C	A,B,C,D	A,C,D	B,C,D	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	4.00	3.00	0.25	0.00	4.00	1.50

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

 1. **Ans (A)**

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{10} - \frac{1}{20} = \frac{1}{20}$$

$$f = +20$$

$$\therefore \frac{1}{v} - \frac{1}{-10} = \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{10} = \frac{-1}{20} = -20\text{cm}$$

$$\therefore m = \frac{-20}{-10} = 2$$

 2. **Ans (C)**

$$E_1 = \frac{1}{2} kx^2$$

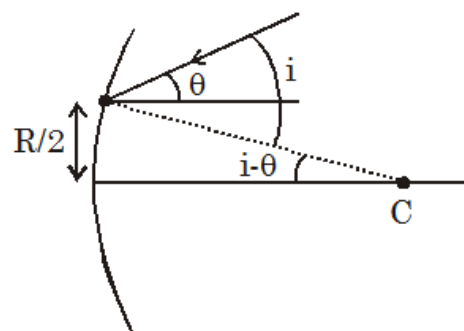
$$E_2 = \frac{1}{2} ky^2$$

$$E = \frac{1}{2} k(x+y)^2$$

 3. **Ans (C)**

The ray will be totally reflected when angle of incidence is equal to critical angle.

$$\frac{\sin i}{\sin \frac{\pi}{2}} = \frac{\mu_2}{\mu_1} \Rightarrow \sin i = \frac{\mu_0}{\mu_1} = \frac{1}{2} \Rightarrow i = 30^\circ$$



From figure, $i - \theta = 30^\circ \Rightarrow \theta = 0$

4. Ans (A)

Both blocks loose contact immediately after the release.

$$T_P = 2\pi\sqrt{\frac{m}{4K}}, T_Q = 2\pi\sqrt{\frac{m}{K}}$$

$$\Rightarrow \therefore T_Q = 2T_P$$

Q comes at lowest position at time $\frac{T_Q}{2}$

travelling a distance $\frac{2mg}{K}$ downwards.

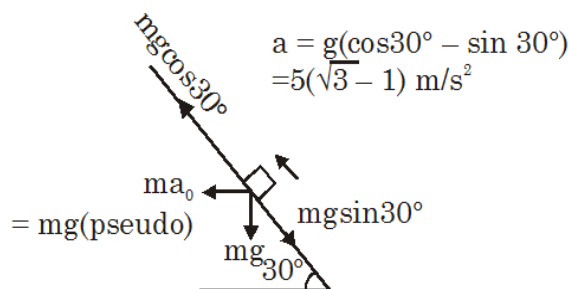
In time $\frac{T_Q}{2}$, i.e. time period of P (T_P) the block

P come back to original position

$\therefore$ The distance between P and Q is $\frac{2mg}{K}$

5. Ans (B)

w.r.t. incline (N.I.F.)



$$\therefore t_{CA} = \sqrt{\frac{2\ell}{a}} = \sqrt{\frac{2}{5(\sqrt{3} - 1)}}$$

$$= \sqrt{\frac{\sqrt{3} + 1}{5}} \approx \sqrt{\frac{2.7}{5}} \approx 0.74 \text{ sec}$$

6. Ans (D)

At $t = 3$

$$\text{Slope} = \frac{d\left(\frac{1}{v}\right)}{dt} = -1$$

$$\Rightarrow -\frac{1}{v^2} \frac{dv}{dt} = -1 \Rightarrow \left(\frac{1}{\sqrt{3}}\right)^2 \frac{dv}{dt} = 1$$

$$\Rightarrow a = \frac{dv}{dt} = 3 \text{ m/s}^2$$

PART-1 : PHYSICS

SECTION-I (ii)

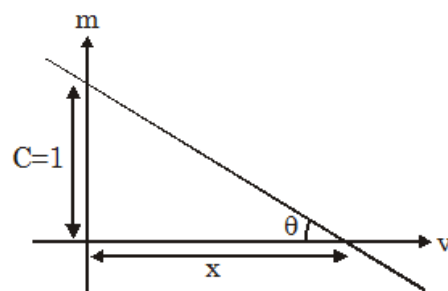
7. Ans (A,B,C)

$$m = \frac{f - v}{f}$$

$$m = 1 - \frac{v}{f}$$

$$\text{Slope} = -\frac{1}{f}$$

$$C = 1$$



$$\tan \theta = \frac{1}{x} = \frac{1}{f}$$

$$x = f$$

8. Ans (A,D)

$$K_{\max} = 16\pi^2 \quad T = 0.2 \text{ s}$$

$$f = \frac{1}{T} = 5 \text{ Hz}$$

$$\frac{1}{2}mv_{\max}^2 = K_{\max} \Rightarrow v_{\max} = 4\pi$$

$$A \frac{2\pi}{0.2} = 4\pi \Rightarrow A = 0.4 \text{ m}$$

At $t = 0.1$ sec the particle is at extreme position so acceleration is not zero.

9. Ans (A,C)

When velocity of object is perpendicular to mirror, relative velocity is maximum and

$$v_{\max} = 2 \text{ m/s.}$$

When velocity of object is parallel to mirror, relative velocity is minimum and $v_{\min} = 0 \text{ m/s}$

10. Ans (A,B,C,D)

$$\therefore F = \frac{-du}{dy} = 0 \quad (\text{For equilibrium})$$

Particle is performing simple harmonic motion

11. **Ans (A,B,D)**

(A, B) is moving with constant velocity then $a = 0$

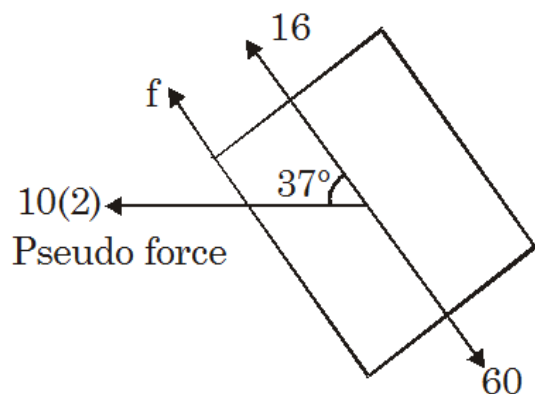
so friction available $= \mu mg \cos \theta$

$$= (0.8)(10)(10)(4/5) = 64 \text{ N.}$$

but $mg \sin \theta = 60 \text{ N}$

so required friction is 60N.

So net force is zero.



$$(C) a = 2\hat{i} \quad f = mg \sin 37^\circ - ma \cos 37^\circ = 44 \text{ N}$$

$$(D) f = mg \sin 37^\circ - ma \cos 37^\circ = -20 \text{ N}$$

12. **Ans (C,D)**

$$\frac{dx}{dt} = 8x + 1$$

$$\int_0^x \frac{dx}{8x+1} = \int_0^1 dt$$

$$\Rightarrow \ln(8x+1) = 8t \Rightarrow x = \frac{1}{8}(e^{8t} - 1) \Rightarrow$$

$$\frac{d^2x}{dt^2} > 0$$

$$\text{at } t = 0, x = 0 \Rightarrow v = 1 \text{ m/s}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (0.50)**

Information based.

2. **Ans (4.00)**

$$T = 2\pi \sqrt{\frac{I}{C}} = 2\pi \sqrt{\frac{2}{5} \frac{mR^2}{C}}$$

$$I = \frac{2}{5} mR^2$$

3. **Ans (1.72 to 1.73)**

$$2(60 - r) = [2(60 - r) + 180 - 2r] \frac{1}{3}$$

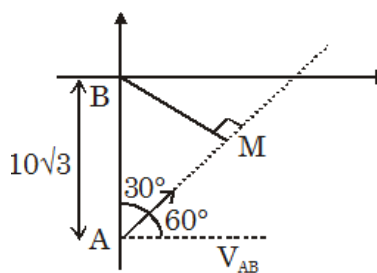
$$6(60 - r) - (60 - r) = 180 - 2r$$

$$4(60 - r) = 180 - 2r$$

$$60 = 2r \Rightarrow r = 30^\circ$$

$$\mu = \sqrt{3}$$

6. **Ans (1.50)**



As observed by B motion of A is along AM and BM is the shortest distance between them.

Relative displacement of A w.r.t to B is

$$AM = AB \cos 30^\circ \quad \vec{V}_{AB} = 5\hat{i} + 5\sqrt{3}\hat{j}$$

Time taken

$$(t) = \left| \frac{\vec{S}_{AB}}{\vec{V}_{AB}} \right| = \frac{AM}{V_{AB}} = \frac{10\sqrt{3} \cos 30^\circ}{10} = 1.5 \text{ sec}$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. **Ans (A)**

More no. of particle, implies more depression in freezing point hence lesser freezing point.

4. **Ans (C)**

$$\text{Net Na}^+ = 4 - \left(2 \times \frac{1}{4} + 1 \right) = 5/2, \text{ Net Cl}^- = 4,$$

$$PF = \frac{\frac{5}{2} \times \frac{4}{3} \pi r_+^3 + 4 \times \frac{4}{3} \pi r_-^3}{16\sqrt{2} r_-^3}$$

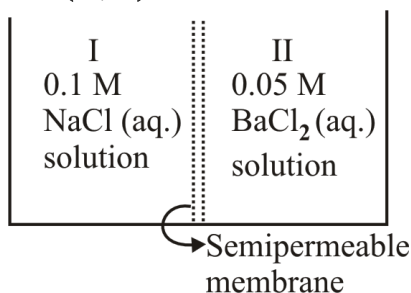
$$\sqrt{2} a = 4r_-$$

$$a = \frac{4r_-}{\sqrt{2}} = 2\sqrt{2} r_-$$

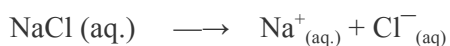
PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (B,D)



In partition I



$$C = 0.1\text{M} \quad 0.1\text{M} \quad 0.1\text{M}$$

$$i = 1 + (n-1) \propto (n = \text{no. of ions})$$

$$i = 1 + (2-1) \times 1 \quad \text{assuming 100\% dissociation}$$

dissociation

$$i = 2 \quad \text{NaCl \& BaCl}_2 \rightarrow \text{strong}$$

electrolyte

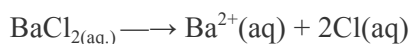
$$\text{So,} \quad \% \alpha = 100\%$$

$$\pi_I = iCRT \quad \boxed{\alpha = 1}$$

$$\pi_I = 2 \times 0.1 \times R \times T$$

$$\pi_I = 0.2 RT$$

In partition II



$$C = 0.05\text{M} \quad 0.05\text{M} \quad 2 \times 0.05\text{M}$$

$$n = 3 \text{ and } \alpha = 1$$

$$i = 1 + (3-1) \times 1 \Rightarrow \boxed{i = 3}$$

$$\text{So, } \pi_{II} = i \times CRT$$

$$= 3 \times 0.05 \times RT$$

$$= 0.15 RT$$

$$\text{So, } \pi_I = 0.20 RT$$

$$\text{and } \pi_{II} = 0.15 RT$$

$$\text{hence } \pi_I > \pi_{II}$$

So, Solution in partition I is hypertonic. Due to that movement of water from BaCl₂ will flow towards the NaCl solution.

(B) option (B) is correct

$$\pi_I (\text{NaCl}) > \pi_{II} (\text{BaCl}_2)$$

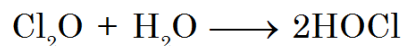
(D) option (D) is correct

$$\text{So, } \boxed{\text{Ans : - (B) \& (D)}}$$

10. Ans (A,B,D)

Part of theory

11. Ans (A,C,D)



12. Ans (A,B,D)

The order of solubility of

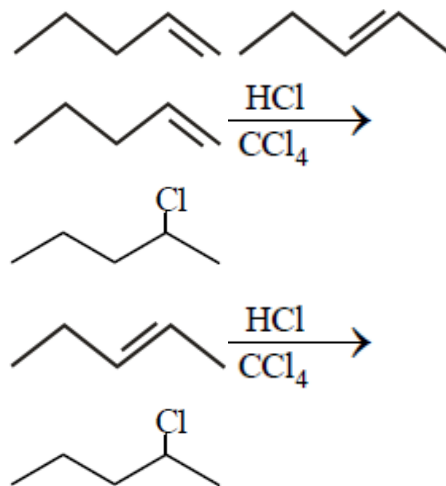


can only be explained on the basis of lattice energy rest others can be explained by Fajan's rule.

PART-2 : CHEMISTRY

SECTION-II

2. Ans (2.00)



3. Ans (135.00)

let mmoles of each is = x

$$\text{n-factor of FeO} = 1$$

$$\text{n-factor of Fe}_{0.80}\text{O} = 0.4$$

$$m_{\text{eq}} \text{ of FeO} + m_{\text{eq}} \text{ of Fe}_{0.80}\text{O} = m_{\text{eq}} \text{ of KMnO}_4$$

$$x \times 1 + x \times 0.4 = 70 \times 0.3 \times 5$$

$$x = 75 \text{ mmoles}$$

mmoles of Fe³⁺ produced

$$= 75 + 75 \times 0.8 = 135 \text{ mmoles}$$

5. **Ans (4.00)**

Refer to notes.

6. **Ans (8.00)**

Refer to notes.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$$\lim_{x \rightarrow \infty} \left(\frac{x^3 + 1}{x^2 + 1} - (ax + b) \right) = 2$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{x^3(1-a) - bx^2 - ax + (1-b)}{x^2 + 1} = 2$$

$$\lim_{x \rightarrow \infty} \frac{x(1-a) - b - \frac{a}{x} + \frac{(1-b)}{x^2}}{1 + \frac{1}{x^2}} = 2$$

$$\Rightarrow 1 - a = 0, -b = 2$$

$$\Rightarrow a = 1, b = -2$$

2. **Ans (A)**

$$\frac{1}{x} = 2 \sin(\cos^{-1} x)$$

$$\frac{1}{x} = 2\sqrt{1-x^2}$$

$$\Rightarrow 4x^4 - 4x^2 + 1 = 0$$

$$\Rightarrow x^2 = \frac{1}{2} \quad (\text{Verify also})$$

3. **Ans (A)**

Function is continuous at $x = -\frac{\pi}{2}$, then

$$f\left(-\frac{\pi}{2}^-\right) = f\left(-\frac{\pi}{2}\right) = f\left(-\frac{\pi}{2}^+\right)$$

$$4 = -a + b \quad \dots(1)$$

function is continuous at $x = \frac{\pi}{2}$

$$f\left(\frac{\pi}{2}^-\right) = f\left(\frac{\pi}{2}\right) = f\left(\frac{\pi}{2}^+\right)$$

$$a + b = 2 \quad \dots(2)$$

from (1) and (2) get a and b

4. **Ans (A)**

$$2\log_e(2^x - 5) = \log_e 2 + \log_e \left(2^x - \frac{7}{2}\right)$$

$$(2^x - 5)^2 = 2 \left(2^x - \frac{7}{2}\right) \quad \text{put } 2^x = t$$

$$t^2 - 10t + 25 = 2t - 7$$

$$t^2 - 12t + 32 = 0$$

$$t = 8, 4 \text{ (rejected)}$$

$$t = 8 = 2^x \Rightarrow x = 3$$

$$\sqrt{3\sqrt{3\sqrt{3\ldots}}} \Rightarrow x = \sqrt{3x} \Rightarrow x = 3$$

5. **Ans (C)**

$$\lim_{x \rightarrow \infty} \frac{x^{1/3} x^{2/3} \left(1 + \frac{1}{x^2}\right)^{1/3}}{(x^2)^{1/2} \left(a + \frac{b}{x^2} + \frac{c}{x^3}\right)^{1/2}} = 2$$

$$\lim_{x \rightarrow \infty} \frac{x}{x} \frac{\left(1 + \frac{1}{x^2}\right)^{1/3}}{\left(a + \frac{b}{x^2} + \frac{c}{x^3}\right)^{1/2}} = 2$$

$$\Rightarrow \frac{1}{a^{1/2}} = 2$$

$$\Rightarrow \frac{1}{2} = \sqrt{a}$$

$$\Rightarrow \boxed{\frac{1}{4} = a}$$

6. **Ans (D)**

$$\text{Let } \frac{1}{x} = y$$

$$\lim_{y \rightarrow 0} \left\{ \frac{(1+y)^{1/y} - e}{y} \right\}$$

$$= \lim_{y \rightarrow 0} \frac{e^{\frac{1}{y} \ln(1+y)} - e}{y}$$

$$= e \lim_{y \rightarrow 0} \frac{e^{\frac{1}{y} \ln(1+y)-1} - 1}{y}$$

$$= e \lim_{y \rightarrow 0} \frac{e^{\frac{1}{y} \ln(1+y)-1} - 1}{\frac{1}{y} \ln(1+y) - 1} \times \frac{\ln(1+y) - y}{y^2}$$

$$= e \lim_{y \rightarrow 0} \frac{\ln(1+y) - y}{y^2} = e \lim_{y \rightarrow 0} \frac{\left(y - \frac{y^2}{2} + \dots\right) - y}{y^2}$$

$$= -\frac{e}{2}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A,B,C)

$$(A) \lim_{x \rightarrow 0} \left[\frac{\sin 2x}{2x} \cdot 2 \right] = [2^-] = 1$$

$$(B) \lim_{x \rightarrow 0} \left[-3 \frac{\tan^{-1} 3x}{3x} \right] = -3$$

$$(C) \left[\lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot 3 \right] = 3$$

$$(D) \lim_{x \rightarrow 0} \left[\frac{\sin^{-1} x}{x} \right] = [1^+] = 1$$

8. Ans (B,C)

$$\lim_{x \rightarrow 0} (e^x + 2x)^{\frac{3}{x}} \quad (1^\infty \text{ form})$$

$$= e^{\lim_{x \rightarrow 0} \frac{(e^x + 2x - 1)3}{x}}$$

$$= e^{3 \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} + 2 \right)} = e^{3(1+2)} = e^9$$

$$\lim_{x \rightarrow \infty} (e^x + 2x)^{\frac{3}{x}} \quad (\infty^0 \text{ form})$$

$$= e^3 \lim_{x \rightarrow \infty} \left(1 + \frac{2x}{e^x} \right)^{\frac{3}{x}} = e^3 (1)^0 = e^3$$

9. Ans (A,B,C,D)

$$(A) f(x) = |x|, g(x) = x \quad \text{NOT Identical}$$

$$(B) f(x) = \sqrt{\frac{x-2}{x-3}}, g(x) = \frac{\sqrt{x-2}}{\sqrt{x-3}}$$

NOT Identical

$$x \in (-\infty, 2) \cup (3, \infty), x \in (3, \infty)$$

$$(C) f(x) = \tan x, g(x) = \frac{1}{\cot x}$$

NOT Identical

$$x \in \mathbb{R} - (2n+1) \frac{\pi}{2},$$

$$x \in \mathbb{R} - \left\{ n\pi, (2n+1) \frac{\pi}{2} \right\}$$

$$(D) f(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}, g(x) = \operatorname{sgn} x$$

NOT Identical

10. Ans (A,C,D)

$$f(x) = \sin \left(\ln \left(x + \sqrt{x^2 + 1} \right) \right)$$

$$\text{Option A : } \ln(x + \sqrt{x^2 + 1}) \rightarrow \text{odd}$$

$$\sin x \rightarrow \text{odd}$$

$$\Rightarrow \sin \left(\ln(x + \sqrt{x^2 + 1}) \right) \text{ is odd.}$$

Option B : odd function is symmetric about origin & not y-axis

$$\text{Option C : } x + \sqrt{x^2 + 1} > 0 \quad \forall x \in \mathbb{R}$$

$$\text{Option D : } x = 0 \Rightarrow f(x) = 0$$

11. Ans (B,C,D)

$$(A) \lim_{x \rightarrow 1^+} [x]^{\frac{1}{x-1}} = (\text{Ex.1})^\infty = 1$$

$$(B) \lim_{x \rightarrow 3} \frac{(x^2 - 9) - |x - 3|}{|x - 1| - 2}$$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(x+3) - |x-3|}{x-3}$$

$$\therefore \lim_{x \rightarrow 3^+} \frac{(x-3)(x+3) - (x-3)}{x-3} = \lim_{x \rightarrow 3^+} x + 3 - 1 = 5$$

$$\lim_{x \rightarrow 3^-} \frac{(x-3)(x+3) + (x-3)}{x-3} = \lim_{x \rightarrow 3^-} x + 3 + 1 = 7$$

DNE

$$(C) \lim_{x \rightarrow 0^+} x^{\ln x} = (0^+)^{-\infty} = \text{DNE}$$

$$(D) \lim_{x \rightarrow 0^+} \left(\frac{1 - \cos(\sin^2 x)}{x^2} \right)^{\frac{\ln(1-2x^2)}{\sin^2 x}} = \lim_{x \rightarrow 0^+} (A)^B$$

$$\lim_{x \rightarrow 0^+} A = \lim_{x \rightarrow 0^+} \frac{1 - \cos(\sin^2 x)}{x^2}$$

$$= \lim_{x \rightarrow 0^+} \frac{1 - \cos(\sin^2 x)}{(\sin^2 x)^2} \cdot \frac{\sin^4 x}{x^2} = 0$$

$$\lim_{x \rightarrow 0^+} B = \lim_{x \rightarrow 0^+} \frac{\ln(1-2x^2)}{\sin^2 x}$$

$$= \lim_{x \rightarrow 0^+} \frac{\ln(1-2x^2)}{-2x^2} \cdot \frac{-2x^2}{\sin^2 x} = -2$$

$$\therefore (0)^{-2} : \text{DNE.}$$

12. Ans (B,C,D)

$$f(x) = \lim_{n \rightarrow \infty} \left(\frac{(1-x)(1+x)(1+x^2) \dots (1+x^{2^{2020}n})}{1-x} \right), |x| < 1$$

$$f(x) = \frac{1}{1-x}$$

$$f\left(\frac{1}{2}\right) = 2, f\left(\frac{1}{3}\right) = \frac{3}{2}$$

$$\sin^{-1} \sin 2 = \pi - 2; \cos^{-1} \cos 2 = 2$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (4.00)**

$$P = \log_4 p_1 + \log_4 p_2 + \log_4 p_3 + \log_4 p_4$$

$$= \log_4 (p_1 p_2 p_3 p_4)$$

$$\text{Also } p_1 + p_2 + p_3 + p_4 \geq 4(p_1 p_2 p_3 p_4)^{1/4}$$

$$\therefore P \leq 4 \Rightarrow N = 4$$

$$\therefore \log_2 N^2 = 4$$

2. **Ans (3.00)**

$$\lim_{x \rightarrow 0} \frac{2 \sin \frac{3x}{2} \cdot \cos \left(\frac{3x+2a}{2} \right) - 3 \left(2 \sin \frac{x}{2} \cdot \cos \frac{3x+2a}{2} \right)}{x^3}$$

$$= \lim_{x \rightarrow 0} 2 \cos \left(\frac{3x+2a}{2} \right) \left(\frac{-4 \left(\frac{x}{2} \right)^3}{x^3} \right)$$

$$= -\cos a$$

3. **Ans (0.25)**

If function is inverse of itself then

$$f(x) = f^{-1}(x)$$

$$\frac{x+a}{x+b} = \frac{a-bx}{x-1}$$

solving, we get $b = -1$

4. **Ans (0.00)**

$f(x)$ is either increasing or decreasing but $f(0) = 5$

$\Rightarrow f(x)$ is decreasing

Hence $f(3) = 0$

5. **Ans (4.00)**

Let $a = \log_5 x$, $x > 1$ so $a > 0$

$$\log_x 0.04 = \log_x 5^{-2} = \frac{-2}{\log_5 x} = \frac{-2}{a}$$

$$\text{Now, } 2 \log_5 x - \log_x 0.04 = 2a + \frac{2}{a}$$

By A.M $\geq$ GM (for $a > 0$)

$$\frac{2a + \frac{2}{a}}{2} \geq \sqrt{2a \times \frac{2}{a}}$$

$$\Rightarrow 2a + \frac{2}{a} \geq 4$$

so least value is 4.

6. **Ans (1.50)**

$$\lim_{x \rightarrow \infty} (g(x) - f(x)) = \frac{\pi}{3}$$

$$\text{or } \left(\sin^{-1}(a) - \sin^{-1} \left(\frac{1}{2} \right) \right) = \frac{\pi}{3}$$

$$\text{or } \sin^{-1}(a) = \frac{\pi}{2} \text{ or } a = 1$$

$$\text{Again } \lim_{x \rightarrow 0} f(x) + g(x) = \frac{\pi}{4}$$

$$\sin^{-1} \left(\frac{3}{5} \right) + \sin^{-1} \left(\frac{b}{5} \right) = \tan^{-1}(1)$$

$$\text{or } \sin^{-1} \left(\frac{b}{5} \right) = \tan^{-1}(1) - \tan^{-1} \left(\frac{3}{4} \right)$$

$$\Rightarrow \sin^{-1} \left(\frac{b}{5} \right) = \sin^{-1} \left(\frac{1}{\sqrt{50}} \right)$$

$$\Rightarrow b = \frac{1}{\sqrt{2}}$$

$$\therefore a^2 + b^2 = 1 + \frac{1}{2} = \frac{3}{2}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,D	A,B,C,D	B,C,D	A,B,C	A,B	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.40	15.00	0.37 to 0.40	25.00	2.50	7.50
SECTION-III	Q.	1	2	3	4	5	6
	A.	6	9	3	7	3	9

PART-2 : CHEMISTRY

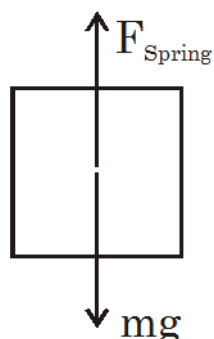
SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,D	A,C,D	A,D	A,B,C	B,C,D	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	6.00	6.00	374.15	2.44 to 2.45	3.00	5.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	1	4	4	2	1	4

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	C,D	B,C	A,B	A,B	A	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.33 to 0.34	6.66 to 6.67	1.00	49.00	3.33 to 3.34	0.20
SECTION-III	Q.	1	2	3	4	5	6
	A.	6	1	4	5	3	2

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (A,B,D)



Speed will be maximum when the block passes through the equilibrium position.

$$F_{\text{spring}} = mg$$

 $\therefore$ A is correct

(B) From conservation of energy

$$mg(\ell + x) = \frac{1}{2}Kx^2 + \frac{1}{2}mv_{\text{max}}^2 \quad \dots (i)$$

$$\text{Where } x = \frac{mg}{K} = \frac{mg}{4mg}\ell = \frac{\ell}{4}$$

$$\text{putting } x = \frac{\ell}{4} \text{ in equation (i) } V_{\text{max}} = \frac{3}{2}\sqrt{g\ell}$$

(C) is incorrect as in equilibrium compression

$$\frac{\ell}{4} \text{ which is more than } \frac{\ell}{8}$$

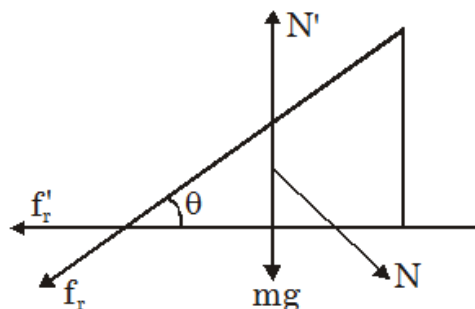
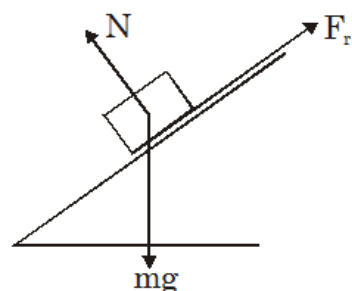
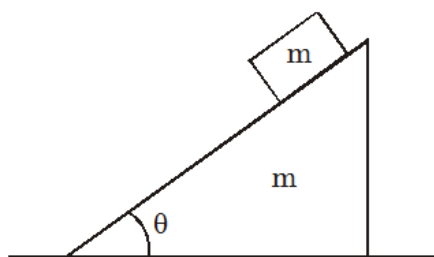
$$(D) \text{ Time taken is } \frac{T}{4} = \frac{1}{4}2\pi\sqrt{\frac{m}{K}}$$

 $\therefore$ D is correct

2. Ans (A,B,C,D)

The above lens mirror combination is an equivalent converging mirror.

3. Ans (B,C,D)

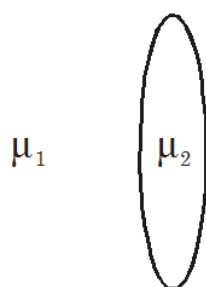


$$N' = mg + N \cos \theta + f_r \sin \theta$$

$$N \sin \theta = f_r' + f_r \cos \theta$$

(for inclined plane to move over horizontal surface)

4. Ans (A,B,C)

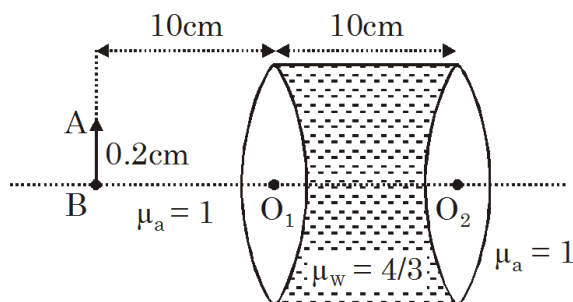


$$\frac{\mu_3}{V} - \frac{\mu_1}{U} = \frac{\mu_3 - \mu_2}{R_2} + \frac{\mu_2 - \mu_1}{R_1}$$

5. Ans (A,B)

As the prism are identical so deviation will be zero for even number of prism and δ for odd number of prism.

6. Ans (A,C)



For the image formed by the first lens

$$\begin{aligned} \frac{\mu_3}{v} - \frac{\mu_1}{u} &= \left(\frac{\mu_2 - \mu_1}{R_1} \right) + \left(\frac{\mu_3 - \mu_2}{R_2} \right) \\ \Rightarrow \frac{4}{3v} - \frac{1}{(-10)} &= \frac{\left(\frac{3}{2} - 1 \right)}{10} + \frac{\left(\frac{4}{3} - \frac{3}{2} \right)}{-10} \\ \frac{4}{3v} + \frac{1}{10} &= \frac{1}{20} + \frac{1}{60} \\ \Rightarrow \frac{4}{3v} &= \frac{1}{15} - \frac{1}{10} \\ \Rightarrow v &= -40 \text{ cm} \end{aligned}$$

Lateral magnification,

$$m_1 = \frac{\mu_1}{\mu_3} \frac{v}{u} = \frac{1 \times 3}{4} \left(\frac{-40}{-10} \right) = +3$$

For the image formed by the second lens

$$\begin{aligned} \frac{\mu_3}{v} - \frac{\mu_1}{u} &= \left(\frac{\mu_2 - \mu_1}{R_1} \right) + \left(\frac{\mu_3 - \mu_2}{R_2} \right) \\ \Rightarrow \frac{1}{v} - \frac{4}{3(-50)} &= \left(\frac{\frac{3}{2} - \frac{4}{3}}{10} \right) + \left(\frac{1 - \frac{3}{2}}{-10} \right) \\ \Rightarrow \frac{1}{v} + \frac{2}{75} &= \frac{1}{60} + \frac{1}{20} \\ \Rightarrow \frac{1}{v} &= \frac{1}{15} - \frac{2}{75} \\ \Rightarrow v &= +25 \text{ cm} \end{aligned}$$

The position of final image formed is 25 cm right to the second lens.

Lateral magnification,

$$m_2 = \frac{\mu_1}{\mu_3} \frac{v}{u} = \frac{4}{3 \times 1} \left(\frac{25}{-50} \right) = -\frac{2}{3}$$

Overall magnification,

$$m = m_1 \times m_2 = (+3) \times \left(-\frac{2}{3} \right) = -2$$

The size of the final image formed = $2 \times 0.2 \text{ cm}$
= 0.4 cm

PART-1 : PHYSICS

SECTION-II

1. **Ans (2.40)**

$$V_x = 180 \cos 60^\circ = 9 \text{ m/s.}$$

$$V_{ly} = \sqrt{15^2 - 9^2} = 12 \text{ m/s}$$

$$V_{2y} = -12 \text{ m/s}$$

$$\therefore \Delta t = \frac{-12 - (12)}{-10} = \frac{12}{5} \text{ sec.}$$

2. **Ans (15.00)**

$$V_B \cos \theta_B = V_A \cos \theta_A \Rightarrow V_B = 15 \text{ m/s}$$

3. **Ans (0.37 to 0.40)**

Applying Snell's law between the points O and

P, we have

$$2 \times \sin 60^\circ = (\sin 90^\circ) \times \frac{2}{(1 + H^2)},$$

$$2 \times \frac{\sqrt{3}}{2} = 1 \times \frac{2}{(1 + H^2)}$$

$$(1 + H^2) = \frac{2}{\sqrt{3}}, H = \sqrt{\left(\frac{2}{\sqrt{3}} - 1\right)}$$

4. **Ans (25.00)**

$$\overrightarrow{PQ} = \sqrt{15^2 + 20^2} = 25 \text{ m}$$

$$25 = \frac{V_p^2 \sin 2(45)}{g} \Rightarrow V_p = 5\sqrt{10} \text{ m/s}$$

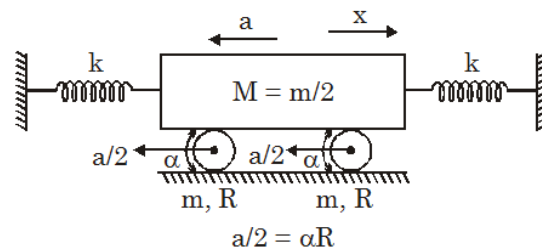
$$\therefore V_A^2 \sin^2 \theta - 2 \times 10 \times 12.5 = (5\sqrt{5})^2$$

$$\Rightarrow V_A \sin \theta = 5\sqrt{15} \text{ m/s}$$

$$V_A \cos \theta = V_p \cos 45^\circ = 5\sqrt{5} \text{ m/s and } \theta = 60^\circ$$

$$\therefore \overrightarrow{AB} = \frac{V_A^2 \sin 2\theta}{g} = 25\sqrt{3}$$

5. **Ans (2.50)**



$$2kx - 2f_1 = Ma = \frac{ma}{2} \dots\dots(i)$$

$$f_1 + f_2 = \frac{ma}{2} \dots\dots(ii)$$

$$(f_1 - f_2)R = \frac{mR^2}{2}\alpha,$$

$$(f_1 - f_2) = \frac{m\alpha R}{2} = \frac{ma}{4} \dots\dots(iii)$$

Adding (i), (ii) and (iii), we get

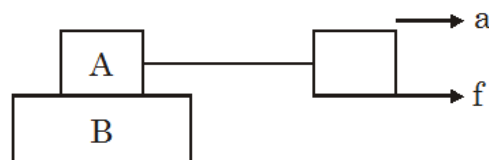
$$2kx = \frac{5ma}{4} \Rightarrow a = \frac{8kx}{5m}$$

$$\frac{d^2x}{dt^2} = -\left(\frac{8k}{5m}\right)x \Rightarrow T = 2\pi\sqrt{\frac{5m}{8k}} = \pi\sqrt{\frac{5m}{2k}}$$

6. **Ans (7.50)**

If all blocks moves with same acceleration

$$a = \frac{F}{2M + 2m}$$



$$f = (m + m + M)a = \frac{(2m + M)F}{2(m + M)}$$

For no slipping $f \leq f_1$

$$\frac{(2m + M)F}{2(m + M)} \leq \mu_s mg$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (6)**

$$\frac{1}{f_L} = \left(\frac{3}{2} - 1\right) \left(\frac{1}{20} + \frac{1}{-20}\right)$$

$$\Rightarrow f_L = 20 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{-10} - \frac{2}{20} = -\frac{1}{5}$$

$$\frac{1}{v} + \frac{1}{-30} = \frac{1}{-5} \Rightarrow \frac{1}{30} - \frac{1}{5}$$

$$\text{For lens } \frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$$

2. Ans (9)

$$a = \mu g$$

$$6^2 = 0^2 + 2 \times \mu g S$$

3. Ans (3)

$$\therefore \mu = \frac{\sin\left(A + \frac{\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \Rightarrow \frac{2}{\sqrt{3}} = \frac{\sin\left(\frac{A + \frac{A}{2}}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \frac{2}{\sqrt{3}} = \frac{\sin\left(\frac{3A}{4}\right)}{\sin\left(\frac{A}{2}\right)}, \text{ Let } \frac{A}{4} = \theta$$

$$\Rightarrow \frac{2}{\sqrt{3}} = \frac{\sin \theta - 4\sin^3 \theta}{2 \sin \theta \cdot \cos \theta} = \frac{3 - 4\sin^2 \theta}{2 \cos \theta}$$

$$\Rightarrow 4\sqrt{3}\cos^2 \theta - 4 \cos \theta - \sqrt{3} = 0$$

$$\Rightarrow \cos \theta = \frac{1 \pm 2}{2\sqrt{3}} = \frac{\sqrt{3}}{2}, \left(-\frac{1}{2\sqrt{3}}\right)$$

$$\Rightarrow \theta = 30^\circ \Rightarrow \frac{A}{4} = 30^\circ = \frac{\pi}{6}$$

$$\Rightarrow A = \frac{2\pi}{3}$$

$$\therefore C = 3$$

PART-2 : CHEMISTRY

SECTION-II

3. Ans (374.15)

$$\Delta T_b = K_b \cdot m = 0.52 \times \frac{\frac{11.7}{58.5} \times 2 + \frac{34.2}{342}}{0.260} = 1K$$

$$\therefore \text{B.P.} = 373.15 + 1 = 374.15 \text{ K.}$$

PART-2 : CHEMISTRY

SECTION-III

3. Ans (4)

$$\frac{r_+}{r_-} = \frac{1.6}{1.864} = 0.858$$

So, it is CsCl type unit cell

$$\text{So } \sqrt{3} a = 2(r_+ + r_-)$$

$$\text{So } a = \frac{2(1.864 + 1.6)}{\sqrt{3}} \text{ \AA} = 2 \times 2 \text{ \AA} = 4 \text{ \AA}$$

4. Ans (2)

2 ppm

$$M_1 V_1 = M_2 V_2$$

$$50 \times M_1 = \frac{1}{50} \times 0.1$$

$$M_1(\text{HCO}_3^-) = \frac{1}{25 \times 10^3}$$

$$\text{m moles of HCO}_3^- = \frac{1}{25 \times 10^3} \times 50 = \frac{1}{500}$$

$$\text{m moles of Ca}^{2+} = \frac{1}{1000}$$

$$\text{m moles of CaCO}_3 = \frac{1}{1000}$$

$$\text{weight of CaCO}_3 = \frac{1}{1000} \times 10^{-3} \times 100 = 10^{-4}$$

$$\text{hardness} = [\text{weight of CaCO}_3 / \text{weight of water} \times 10^6]$$

$$= \frac{10^{-4}}{50} \times 10^6 = 2 \text{ ppm}$$

5. Ans (1)



6. Ans (4)



PART-3 : MATHEMATICS

SECTION-I

1. Ans (C,D)

$$f(f(x)) = \frac{2f(x) + 1}{f(x)} = \frac{\frac{2(2x+1)}{x} + 1}{\frac{2x+1}{x}}$$

$$= \frac{5x + 2}{2x + 1}$$

$$g(x) = f(f(x)) - \frac{1}{f(x)}, x \neq 0, -\frac{1}{2}$$

$$g(x) = \frac{5x + 2}{2x + 1} - \frac{x}{2x + 1}$$

$$g(x) = 2$$

$\Rightarrow g(x)$ is a constant function but not

symmetrically defined about y-axis

2. **Ans (B,C)**

$$y = \cos^{-1}x + \cot^{-1}x$$

Domain of y is $x \in [-1, 1]$

$\therefore$ both $\cos^{-1}x$ & $\cot^{-1}x$ are decreasing

$$\Rightarrow y_{\max} = \cos^{-1}(-1) + \cot^{-1}(-1)$$

$$= \pi + \frac{3\pi}{4} = \frac{7\pi}{4}$$

$$y_{\min} = \cos^{-1}(1) + \cot^{-1}(1) = 0 + \frac{\pi}{4}$$

$$\Rightarrow y \in \left[\frac{\pi}{4}, \frac{7\pi}{4} \right]$$

$\Rightarrow$ (B), (C) are correct.

3. **Ans (A,B)**

$$\lim_{x \rightarrow 5} \frac{\ln(3f(x) - 5)}{(f(x) - 2)}$$

$$= \lim_{x \rightarrow 5} \frac{\ln(1 + (3f(x) - 6))}{3f(x) - 6} \left(\frac{3f(x) - 6}{f(x) - 2} \right)$$

Given $f(x) \rightarrow 2$ as $x \rightarrow 5 \Rightarrow 3f(x) - 5 \rightarrow 1$

$$\Rightarrow \lim_{x \rightarrow 5} \frac{3f(x) - 6}{f(x) - 2} = 3$$

4. **Ans (A,B)**

Put $x = 1 + z$

$$\lim_{z \rightarrow 0} \frac{e^z - 1 - z}{z^n} = \lim_{z \rightarrow 0} \frac{1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots - 1 - z}{z^n}$$

$$= \lim_{z \rightarrow 0} \frac{z^2 \left(\frac{1}{2!} + \frac{z}{3!} + \dots \right)}{z^n}$$

$\therefore$ Possible values of 'n' among options are 1, 2

5. **Ans (A)**

$$\text{LHL} = \lim_{x \rightarrow 0^-} \cot^2 x \cdot \ln \cos x$$

$$= \frac{a \ln(1 + (\cos x - 1))}{\tan^2 x} = \frac{a(\cos x - 1)}{x^2}$$

$$= -\frac{a}{2}$$

$$f(0) = 1$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} (\sec x)^{b \cot x} = e^{(\sec x - 1)b \frac{\cos x}{\sin x}}$$

$$= e^{\frac{b(1 - \cos x) \cdot x}{x^2}} = e^0 = 1$$

$$\Rightarrow a = -2 \quad \& \quad b \in \mathbb{R}$$

6. **Ans (B,C,D)**

$$b = \log_2 3 \times \log_3 4 \times \log_4 5 \times \dots \times \log_{15} 16$$

$$\Rightarrow b = \log_2 16 \Rightarrow b = 4 \quad \& \quad a = 2^b \Rightarrow a = 16$$

$$\text{Hence } \log_2(\log_{16} a) = 0$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (0.33 to 0.34)**

$$f(x) = \lim_{t \rightarrow 0} \tan^{-1} \left(\frac{(e^{xt} - 1)}{tx} \times x \right) = \tan^{-1} x$$

$$\lim_{x \rightarrow 0} \frac{x - f(x)}{x^3} = \lim_{x \rightarrow 0} \frac{x - \tan^{-1} x}{x^3} = \frac{1}{3}$$

2. **Ans (6.66 to 6.67)**

$$(1)^\infty \text{ form; use } \lim_{x \rightarrow 0} e^{\frac{4}{5x^2} (a \sin bx^2 + 4(1 - \cos^2 x))} = e^2$$

$$\Rightarrow a \frac{\sin bx^2}{x^2} \times \frac{4}{5} + \frac{16}{5} \frac{\sin^2 x}{x^2} = 2$$

$$\frac{4ab}{5} + \frac{16}{5} = 2 \Rightarrow 4ab = -6 \Rightarrow ab = -\frac{2}{3}$$

3. **Ans (1.00)**

$$x = \tan^{-1} 1 + \tan^{-1} 11 + \frac{9\pi}{2}$$

$$x = \frac{9\pi}{2} + \tan^{-1} \left(\frac{1 + 11}{1 - 11} \right) + \pi$$

$$x = \frac{11\pi}{2} - \tan^{-1}(1.2)$$

$$x = 5\pi + \cot^{-1}(1.2)$$

$$\tan = \frac{1}{1.2} = \frac{5}{6}$$

$$p = 5, q = 6$$

4. **Ans (49.00)**

$$S = (2 - d) \left(\frac{2}{3} \right) + (2 + d) \left(\frac{4}{9} \right) + (2 + 3d) \left(\frac{8}{27} \right) + \dots$$

$$S \left(\frac{2}{3} \right) = (2 - d) \frac{4}{9} + (2 + d) \frac{8}{27} + \dots$$

Subtracting

$$\frac{S}{3} = (2 - d) \frac{2}{3} + 2d \left(\frac{4}{9} \right) + 2d \left(\frac{8}{27} \right) + \dots$$

$$= (2 - d) \frac{2}{3} + 2d \left[\frac{4}{9} + \frac{8}{27} + \dots \right]$$

$$= (2 - d) \frac{2}{3} + 2d \left[\frac{4}{9} \cdot \left(\frac{3}{1} \right) \right] = \frac{4}{3} + 2d$$

$$\Rightarrow S = 4 + 6d$$

$$\text{Given } S = \frac{5}{2} - 2 = \frac{1}{2} = 4 + 6d$$

$$\Rightarrow d = -\frac{7}{12}$$

5. **Ans (3.33 to 3.34)**

Put $x = y = 1$

$$f(1) = 2f(2) = 1$$

Now, put $x = 1, y = 2$

$$\Rightarrow 2f(2) = 3f(3) \Rightarrow f(3) = \frac{1}{3}$$

6. **Ans (0.20)**

$$x^{\log_{\sqrt{x}}(x-3)} = 16 \Rightarrow (x-3)^4 = 16$$

$$\Rightarrow (x-3)^2 = 4 \Rightarrow x-3 = \pm 2$$

$$\therefore x = 5, 1$$

but $x \neq 1$

$$\therefore x = 5$$

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (6)**

$$\frac{a+b}{2} = \sqrt{ab} + \frac{3}{2}$$

$$\Rightarrow a+b = 2\sqrt{ab} + 3 \quad \dots\dots(1)$$

$$\& \sqrt{ab} - \frac{2ab}{a+b} = \frac{6}{5}$$

$$\Rightarrow 5\sqrt{ab}(a+b) = 10ab + 6(a+b) \quad \dots\dots(2)$$

put value of $a+b$ in equation (2) from equation (1) then $\sqrt{ab} = 6$

$$\Rightarrow ab = 36$$

$$\therefore a+b = 15$$

$$\Rightarrow a = 12 \& b = 3 \text{ or } a = 3 \& b = 12$$

$$\Rightarrow \alpha = 15 \& \beta = 9$$

2. **Ans (1)**

$$5^{\log_5 x \cdot \log_5 x} + x^{\log_5 x} = 1250$$

$$\Rightarrow (5^{\log_5 x})^{\log_5 x} + x^{\log_5 x} = 1250$$

$$\Rightarrow x^{\log_5 x} + x^{\log_5 x} = 1250 \Rightarrow 2x^{\log_5 x} = 1250$$

$$\Rightarrow x^{\log_5 x} = 625 \Rightarrow \log_5 x^{\log_5 x} = \log_5 625$$

$$\Rightarrow (\log_5 x)(\log_5 x) = 4 \Rightarrow (\log_5 x)^2 = 4$$

$$\Rightarrow \log_5 x = 2 \text{ or } \log_5 x = -2 \Rightarrow x = 25 \text{ or } x = \frac{1}{25}$$

$$\text{So, } \log_{25} \left(\frac{1}{25} \right) = -1$$

3. **Ans (4)**

$$f(x) = \sin^{-1}[2x] + \cos^{-1}([x] - 1)$$

$$-1 \leq [2x] \leq 1 \quad \& \quad -1 \leq [x] - 1 \leq 1$$

$$\Rightarrow -1 \leq 2x < 2 \quad \& \quad 0 \leq [x] \leq 2$$

$$\Rightarrow -\frac{1}{2} \leq x < 1 \quad \& \quad 0 \leq x < 3$$

$$\Rightarrow 0 \leq x < 1 \quad \text{Domain}$$

$$\Rightarrow [x] = 0 \Rightarrow 0 \leq 2x < 2$$

$$\Rightarrow [2x] = 0 \text{ or } 1$$

$$\text{Now } f(x) = \sin^{-1}[2x] + \cos^{-1}(-1)$$

$$= \left(0 \text{ or } \frac{\pi}{2} \right) + \pi = \pi \text{ or } \frac{3\pi}{2}$$

$$\Rightarrow a + b + \frac{2d}{c} = 1 + 3 = 4$$

4. **Ans (5)**

$$f(x) = \begin{cases} \frac{a \sin x + b \tan x - 3x}{x^3}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$a \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} \right) + b \left(x + \frac{x^3}{3} + \frac{2x^5}{15} \right) - 3x$$

$$\lim_{x \rightarrow 0} \frac{\dots}{x^3}$$

$$\text{Limit exist if } a + b - 3 = 0 \quad \dots\dots(i)$$

$$\text{and } \lim_{x \rightarrow 0} f(x) = f(0) = 0$$

$$-\frac{a}{6} + \frac{b}{3} = 0 \quad \dots\dots(ii)$$

From (i) and (ii)

$$a = 2, b = 1 \quad \therefore a^2 + b^2 = 5$$

5. **Ans (3)**

$$A = \frac{a+b}{2}, G = \sqrt{ab} \quad \& \quad H = \frac{2ab}{a+b}$$

$$\text{Now } A = 3H$$

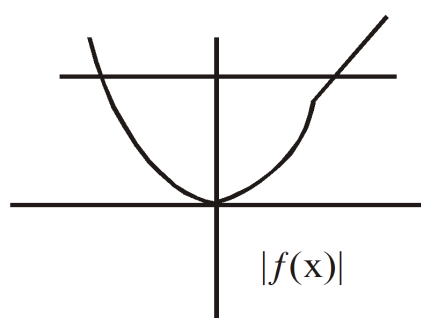
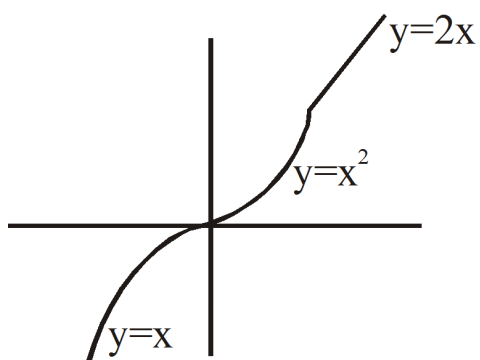
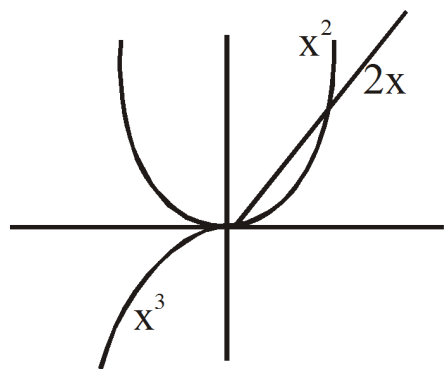
$$\Rightarrow \frac{a+b}{2} = \frac{6ab}{a+b}$$

$$\Rightarrow (a+b)^2 = 12ab$$

$$\Rightarrow \frac{\left(\frac{a+b}{2} \right)^2}{(\sqrt{ab})^2} = 3$$

$$\Rightarrow \frac{A^2}{G^2} = 3$$

6. Ans (2)



$$f(x) = \begin{cases} 2x & ; \quad x \geq 2 \\ x^2 & ; \quad 0 \leq x \leq 2 \\ x^3 & ; \quad x \leq 0 \end{cases}$$

Two solutions.